

main.sh executed in terminal

```
[jiwoov@localhost RailwaySystem]$ ./main.sh
```

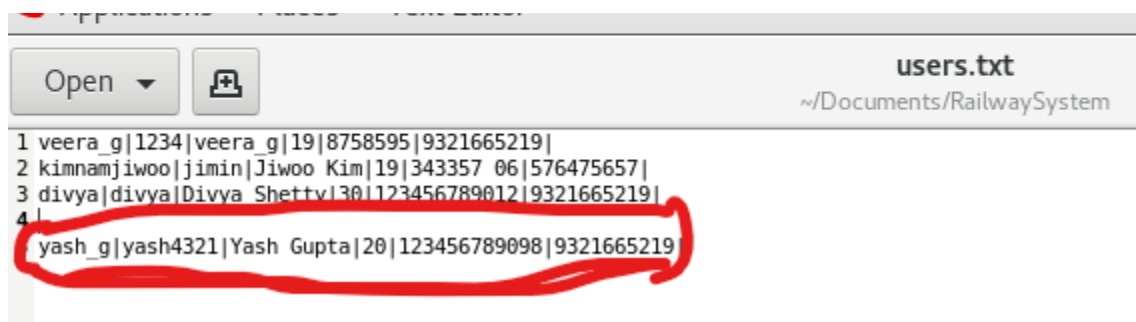
```
=====
RAILWAY TICKET BOOKING SYSTEM
=====
```

1. Register Passenger
2. Passenger Login
3. Check Trains (Guest Mode)
4. Exit

Enter choice:

1. Register passenger

```
Enter choice: 1
===== Passenger Registration Process =====
Name: Yash Gupta
Age: 20
Aadhar Number: 123456789098
Phone Number: 9321665219
Username: yash_g
Password:
Registration successful.
```



Yash Gupta new user added in users.txt

If the username of Yash Gupta is registered again, then "User already exists!" will be shown

```
Enter choice: 1
===== Passenger Registration Process =====
Name: Yash Gupta
Age: 20
Aadhar Number: 123456789098
Phone Number: 9321665219
Username: yash_g
Password:
Username yash_g already exists!
```

2. Passenger Login
Wrong password = Login fail (here I entered incorrect password)

```
Enter choice: 2
===== Passenger Login =====
Username: yash_g
Password:
Passenger not found. Registration required!
```

This time: username is incorrect, password is correct = Login failed

```
Enter choice: 2
===== Passenger Login =====
Username: yash
Password:
Passenger not found. Registration required!
```

Correct Username, Password = Login successful

```
=====
          RAILWAY TICKET BOOKING SYSTEM
=====
1. Register Passenger
2. Passenger Login
3. Check Trains (Guest Mode)
4. Exit

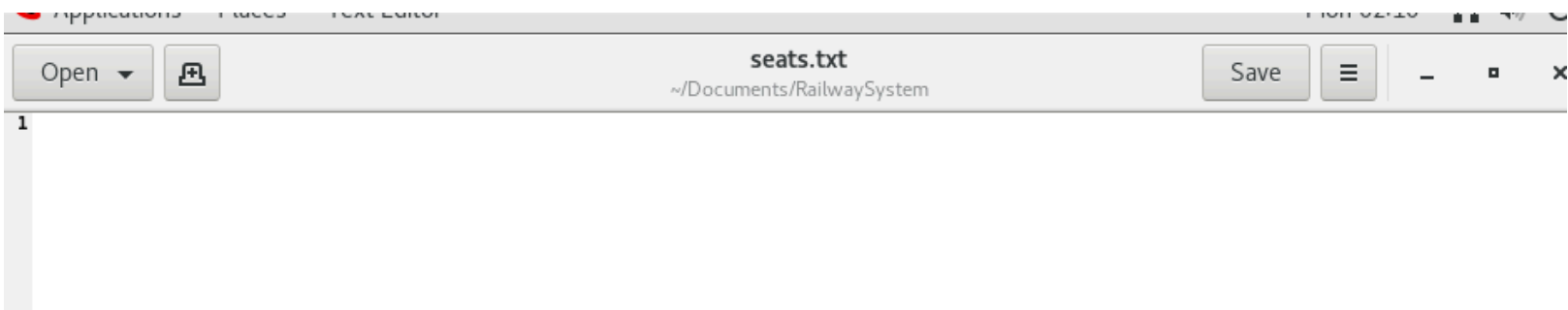
Enter choice: 2
===== Passenger Login =====
Username: yash_g
Password:
Login successful!
```

3. Logout

```
=====
                Passenger Dashboard
=====
Logged in as: yash_g

1. Book Ticket
2. Show my Tickets
3. Cancel Ticket
4. Logout

Kindly enter your choice: 4
Logged out succesfully!
```



seats.txt is currently empty

```
jiwoov@localhost:~/Documents/RailwaySystem
File Edit View Search Terminal Help
[jiwoov@localhost ~]$ cd ~/Documents/RailwaySystem
[jiwoov@localhost RailwaySystem]$ ./generate_seats.sh
Seat data generated successfully!
```

generate_seats.sh executed in terminal

Applications | Files | Text Editor

Open ▾  **seats.txt**
~/Documents/RailwaySystem

```
17538 102|2026-03-07|IAC|H2|43|UB|Available
17539 102|2026-03-07|IAC|H2|46|UB|Available
17540 102|2026-03-07|IAC|H2|47|UB|Available
17541 102|2026-03-07|IAC|H2|48|UB|Available
17542 102|2026-03-07|IAC|H2|49|UB|Available
17543 102|2026-03-07|IAC|H2|50|UB|Available
17544 102|2026-03-07|IAC|H2|51|UB|Available
17545 102|2026-03-08|SL|S1|1|LB|Available
17546 102|2026-03-08|SL|S1|2|LB|Available
17547 102|2026-03-08|SL|S1|3|LB|Available
17548 102|2026-03-08|SL|S1|4|LB|Available
17549 102|2026-03-08|SL|S1|5|LB|Available
17550 102|2026-03-08|SL|S1|6|LB|Available
17551 102|2026-03-08|SL|S1|7|LB|Available
17552 102|2026-03-08|SL|S1|8|LB|Available
17553 102|2026-03-08|SL|S1|9|LB|Available
17554 102|2026-03-08|SL|S1|10|LB|Available
17555 102|2026-03-08|SL|S1|11|LB|Available
17556 102|2026-03-08|SL|S1|12|LB|Available
17557 102|2026-03-08|SL|S1|13|LB|Available
17558 102|2026-03-08|SL|S1|14|LB|Available
17559 102|2026-03-08|SL|S1|15|LB|Available
17560 102|2026-03-08|SL|S1|16|LB|Available
17561 102|2026-03-08|SL|S1|17|LB|Available
17562 102|2026-03-08|SL|S1|18|MB|Available
17563 102|2026-03-08|SL|S1|19|MB|Available
17564 102|2026-03-08|SL|S1|20|MB|Available
17565 102|2026-03-08|SL|S1|21|MB|Available
17566 102|2026-03-08|SL|S1|22|MB|Available
17567 102|2026-03-08|SL|S1|23|MB|Available
17568 102|2026-03-08|SL|S1|24|MB|Available
17569 102|2026-03-08|SL|S1|25|MB|Available
17570 102|2026-03-08|SL|S1|26|MB|Available
17571 102|2026-03-08|SL|S1|27|MB|Available
17572 102|2026-03-08|SL|S1|28|MB|Available
17573 102|2026-03-08|SL|S1|29|MB|Available
17574 102|2026-03-08|SL|S1|30|MB|Available
17575 102|2026-03-08|SL|S1|31|MB|Available
17576 102|2026-03-08|SL|S1|32|MB|Available
17577 102|2026-03-08|SL|S1|33|MB|Available
17578 102|2026-03-08|SL|S1|34|MB|Available
17579 102|2026-03-08|SL|S1|35|UB|Available
17580 102|2026-03-08|SL|S1|36|UB|Available
17581 102|2026-03-08|SL|S1|37|UB|Available
17582 102|2026-03-08|SL|S1|38|UB|Available
17583 102|2026-03-08|SL|S1|39|UB|Available
17584 102|2026-03-08|SL|S1|40|UB|Available
17585 102|2026-03-08|SL|S1|41|UB|Available
17586 102|2026-03-08|SL|S1|42|UB|Available
17587 102|2026-03-08|SL|S1|43|UB|Available
```

1 month's data for tickets will be generated and get stored in **seats.txt**

Open ▾  **trains.txt**
~/Documents/RailwaySystem

```
1 101|Rajdhani Express|Delhi|Mumbai
2 102|Shatabdi Express|Delhi|Chandigarh
3 103|Duronto Express|Kolkata|Delhi
4 104|Garib Rath|Mumbai|Ahmedabad
```

trains.txt . Contains this few trains only for now

1. Check Trains (date working)

According to system - current date - February 2026

If we enter a later date, it will give an error.

```
=====
RAILWAY TICKET BOOKING SYSTEM
=====
```

1. Register Passenger
2. Passenger Login
3. Check Trains (Guest Mode)
4. Exit

Enter choice: 3

===== CHECK ALL AVAILABLE TRAINS FOR YOUR JOURNEY =====

Kindly enter Journey Date (YYYY-MM-DD) within 30 days: 2026-04-01
Invalid date. Kindly select a date that falls within the next 30 days.

Dates within one month are considered valid; beyond that, the train-checking process will be triggered

```
=====
RAILWAY TICKET BOOKING SYSTEM
=====
```

1. Register Passenger
2. Passenger Login
3. Check Trains (Guest Mode)
4. Exit

Enter choice: 3

===== CHECK ALL AVAILABLE TRAINS FOR YOUR JOURNEY =====

Kindly enter Journey Date (YYYY-MM-DD) within 30 days: 2026-03-01

===== SEARCH TRAINS =====

1. By Train Number
2. By Train Name
3. From → To
4. Go Back

Choose option:

2. Check Train by Number

If wrong train number is entered, no data shown

===== SEARCH TRAINS =====

1. By Train Number
2. By Train Name
3. From → To
4. Go Back

Choose option: 1

Enter Train Number: 111

No data

Kindly Press Enter to continue...

Correct Train no. entered

===== SEARCH TRAINS =====

1. By Train Number
2. By Train Name
3. From → To
4. Go Back

Choose option: 1

Enter Train Number: 101

101|Rajdhani Express|Delhi|Mumbai

1AC : 102 seats available

SL : 102 seats available

2AC : 102 seats available

3AC : 102 seats available

1. Book this train
2. Go Back

Kindly enter your chosen option: █

3. Check Train by Name

Provide an input name that is not listed in train.txt. Therefore no data shown

===== SEARCH TRAINS =====

- 1. By Train Number**
- 2. By Train Name**
- 3. From → To**
- 4. Go Back**

Choose option: 2

Enter Train Name: Darbhanga Express

No data

Kindly Press Enter to continue...

On successful name validation, available seats are returned/displayed

===== SEARCH TRAINS =====

- 1. By Train Number**
- 2. By Train Name**
- 3. From → To**
- 4. Go Back**

Choose option: 2

Enter Train Name: Durgam Express

103|Durgam Express|Kolkata|Delhi

1AC : 102 seats available

SL : 102 seats available

2AC : 102 seats available

3AC : 102 seats available

- 1. Book this train**
- 2. Go Back**

Kindly enter your chosen option: █

4. Check trains by from → to

Entered destination pattern doesn't exist in the train list, hence, no data shown

===== SEARCH TRAINS =====

1. By Train Number
2. By Train Name
3. From → To
4. Go Back

Choose option: 3

From: Mumbai

To: Delhi

No data

Delhi → Mumbai destination exist in trains.txt, so seats are shown

===== SEARCH TRAINS =====

1. By Train Number
2. By Train Name
3. From → To
4. Go Back

Choose option: 3

From: Delhi

To: Mumbai

101|Rajdhani Express|Delhi|Mumbai

1AC : 102 seats available

SL : 102 seats available

2AC : 102 seats available

3AC : 102 seats available

1. Book this train

2. Go Back

Kindly enter your chosen option: █

Selecting option 1 (BOOK THIS TRAIN'S TICKET) triggers a login prompt, if the user is not already logged in, since the script booking.sh is called in the code

1. Book this train

2. Go Back

Kindly enter your chosen option: 1

Login required!

===== Passenger Login =====

Username: █

Option 1 . BOOK THIS TRAIN'S TICKET chosen here

===== SEARCH TRAINS =====

- 1. By Train Number**
- 2. By Train Name**
- 3. From → To**
- 4. Go Back**

Choose option: 1

Enter Train Number: 101

101|Rajdhani Express|Delhi|Mumbai

1AC : 102 seats available

SL : 102 seats available

2AC : 102 seats available

3AC : 102 seats available

1. Book this train

2. Go Back

Kindly enter your chosen option: 1

===== BOOK TICKET =====

Select Coach Type (SL/3AC/2AC/1AC):

First, the system will ask for coach type

1. If an invalid coach name is entered, it will show "Invalid."

Kindly enter your chosen option: 1

===== BOOK TICKET =====

Select Coach Type (SL/3AC/2AC/1AC): DS

Invalid coach type!

Kindly Press Enter to continue...

2. If the passenger count entered is incorrect, it will also show "Invalid."

Kindly enter your chosen option: 1

===== BOOK TICKET =====

Select Coach Type (SL/3AC/2AC/1AC): sl

Kindly enter no. of passengers travelling (1-6): 8

Invalid passenger count entered!

Kindly Press Enter to continue...

3. The name field cannot be left empty.

1. Book this train

2. Go Back

Kindly enter your chosen option: 1

===== BOOK TICKET =====

Select Coach Type (SL/3AC/2AC/1AC): SL

Kindly enter no. of passengers travelling (1-6): 2

----- Passenger 1 Details -----

Name:

Name cannot be empty!

Kindly Press Enter to continue...

4. Age validation is applied (allowed range: 0–120).

===== BOOK TICKET =====

Select Coach Type (SL/3AC/2AC/1AC): sl

Kindly enter no. of passengers travelling (1-6): 2

----- Passenger 1 Details -----

Name: Veera Gupta

Age: 300

Invalid age entered!

5. For gender, only M/F is accepted; any other input will be invalid.

===== BOOK TICKET =====

Select Coach Type (SL/3AC/2AC/1AC): SL

Kindly enter no. of passengers travelling (1-6): 2

----- Passenger 1 Details -----

Name: Veera Gupta

Age: 19

Gender (M/F/O): L

Invalid gender! Enter M or F or O.

Kindly Press Enter to continue...

6. Aadhaar number must be exactly 12 digits, otherwise invalid.

----- Passenger 1 Details -----

Name: Veera Gupta

Age: 19

Gender (M/F/O): F

Aadhar No. (12 digits): 1234567

Invalid Aadhar Number!

7. Phone number must be exactly 10 digits.

----- Passenger 1 Details -----

Name: Veera Gupta

Age: 19

Gender (M/F/O): F

Aadhar No. (12 digits): 312757160108

Phone No. (10 digits): 93216652199

Invalid Phone Number!

8. Berth preference must match the options given in brackets; otherwise invalid.

----- Passenger 1 Details -----

Name: Veera Gupta

Age: 19

Gender (M/F/O): F

Aadhar No. (12 digits): 312757160108

Phone No. (10 digits): 9321665219

Berth Preference (LB/MB/UB): PP

Invalid berth preference!

→ WAITING TICKET

Choose option: 1

Enter Train Number: 101

101|Rajdhani Express|Delhi|Mumbai

1AC : 102 seats available

SL : No seats available

2AC : 102 seats available

3AC : 102 seats available

1. Book this train

2. Go Back

Kindly enter your chosen option: 1

===== BOOK TICKET =====

Select Coach Type (SL/3AC/2AC/1AC): SL

Kindly enter no. of passengers travelling (1-6): 1

----- Passenger 1 Details -----

Name: Ranjana Alagi

Age: 18

Gender (M/F/O): F

Aadhar No. (12 digits): 789654123698

Phone No. (10 digits): 9076404130

Berth Preference (LB/MB/UB): LB

No seats available. Ticket will be WAITING!

Do you want to continue booking a WAITING LIST ticket? (Y/N): Y

=====

Total fare: Rs. 500

Seats will be locked for 120 seconds. Complete payment to confirm booking.

1. Proceed to Payment

2. Cancel Booking

Kindly enter your choice:

→ Additional case: If the user selects LB (Lower Berth) but LB is not available, and UB/MB are available, then whichever (UB/MB) appears first in the list will be booked automatically.

Total fare will be auto-calculated (costs are predefined for all coach types).

```
case "$coach_type" in
    SL) fare_per_person=500 ;;
    3AC) fare_per_person=900 ;;
    2AC) fare_per_person=1300 ;;
    1AC) fare_per_person=2000 ;;
esac
```

1. If the UPI ID does not contain the '@' symbol, it will be marked incorrect.

=====

Total fare: Rs. 1000

Seats will be locked for 120 seconds. Complete payment to confirm booking.

1. Proceed to Payment

2. Cancel Booking

Kindly enter your choice: 1

Select Payment Method:

1. UPI

2. Card

Kindly enter your choice: 1

Kindly enter UPI ID: veeraaksbi

Invalid UPI ID. Payment Failed!

Releasing locked seats...

Booking Failed!

Kindly Press Enter to continue...

2. For card payments: Card number must be 16 digits, CVV must be 3 digits.

Select Payment Method:

1. UPI

2. Card

Kindly enter your choice: 2

Kindly enter 16-digit Card Number: 78965215

Kindly enter 3-digit CVV: 125

Invalid Card Details. Payment Failed!

Releasing locked seats...

Booking Failed!

■

Select Payment Method:

1. UPI

2. Card

Kindly enter your choice: 2

Kindly enter 16-digit Card Number: 7852369547412532

Kindly enter 3-digit CVV: 78

Invalid Card Details. Payment Failed!

Releasing locked seats...

Booking Failed!

■

.. ' " ~ -

Only when all information is correct and payment is successful, the ticket will be booked

Aadhar No. (12 digits): 785412693258

Phone No. (10 digits): 2156978541

Berth Preference (LB/MB/UB): UB

=====

Total fare: Rs. 1000

Seats will be locked for 120 seconds. Complete payment to confirm booking.

1. Proceed to Payment

2. Cancel Booking

Kindly enter your choice: 1

Select Payment Method:

1. UPI

2. Card

Kindly enter your choice: 1

Kindly enter UPI ID: veera@oksbi

=====

Ticket Booked Successfully!

PNR: 126-104581

Coach Type: SL

Journey Date: 2026-03-01

Total Passengers: 2

Total Fare: Rs. 1000

=====

Applications Places Text Editor

Open ▾



tickets.txt

~/Documents/RailwaySystem

1 126-104581|yash_g|101|2026-03-01|1771851511|1000|SL-S1-1-LB-CON-Veera Gupta-19-F-312757160108-9321665219,SL-S1-35-UB-CON-Yash Gupta-14-M-785412693258-2156978541|

101	2026-03-01	SL	S1	1	LB	Booked
101	2026-03-01	SL	S1	2	LB	Available
101	2026-03-01	SL	S1	3	LB	Available
101	2026-03-01	SL	S1	4	LB	Available
101	2026-03-01	SL	S1	5	LB	Available
101	2026-03-01	SL	S1	6	LB	Available
101	2026-03-01	SL	S1	7	LB	Available
101	2026-03-01	SL	S1	8	LB	Available
101	2026-03-01	SL	S1	9	LB	Available
101	2026-03-01	SL	S1	10	LB	Available
101	2026-03-01	SL	S1	11	LB	Available
101	2026-03-01	SL	S1	12	LB	Available
101	2026-03-01	SL	S1	13	LB	Available
101	2026-03-01	SL	S1	14	LB	Available
101	2026-03-01	SL	S1	15	LB	Available
101	2026-03-01	SL	S1	16	LB	Available
101	2026-03-01	SL	S1	17	LB	Available
101	2026-03-01	SL	S1	18	MB	Available
101	2026-03-01	SL	S1	19	MB	Available
101	2026-03-01	SL	S1	20	MB	Available
101	2026-03-01	SL	S1	21	MB	Available
101	2026-03-01	SL	S1	22	MB	Available
101	2026-03-01	SL	S1	23	MB	Available
101	2026-03-01	SL	S1	24	MB	Available
101	2026-03-01	SL	S1	25	MB	Available
101	2026-03-01	SL	S1	26	MB	Available
101	2026-03-01	SL	S1	27	MB	Available
101	2026-03-01	SL	S1	28	MB	Available
101	2026-03-01	SL	S1	29	MB	Available
101	2026-03-01	SL	S1	30	MB	Available
101	2026-03-01	SL	S1	31	MB	Available
101	2026-03-01	SL	S1	32	MB	Available
101	2026-03-01	SL	S1	33	MB	Available
101	2026-03-01	SL	S1	34	MB	Available
101	2026-03-01	SL	S1	35	UB	Booked

1. show tickets :

```
=====
Passenger Dashboard
=====
```

Logged in as: yash_g

1. Book Ticket
2. Show my Tickets
3. Cancel Ticket
4. Logout

Kindly enter your choice: 2

===== MY TICKETS =====

- ```

```
1. 2026-03-01 | Train 101 - Rajdhani Express | Delhi → Mumbai
  2. 2026-02-28 | Train 103 - Duronto Express | Kolkata → Delhi

Enter ticket number to view details (0 to go back): █

2. To view a ticket in detail (this will be the final ticket)

Kindly enter your choice: 2

===== MY TICKETS =====

- ```
-----
```
1. 2026-03-01 | Train 101 - Rajdhani Express | Delhi → Mumbai
 2. 2026-02-28 | Train 103 - Duronto Express | Kolkata → Delhi

Enter ticket number to view details (0 to go back): 2

INDIAN RAILWAYS
VeRaSSS Rail

PNR NO : 208-122654
Train No : 103
Train Name : Durlonto Express
From : Kolkata
To : Delhi
Journey Date : 2026-02-28
Booking Time : 23-02-2026 05:10 AM

COACH	SEAT/BERTH/STATUS	SEX	AGE
H1	35/UB/CON	F	18

Fare Paid : ₹2000

Have a Safe Journey & Happy Journey!

This is our ticket format

3. Cancel ticket

Passenger Dashboard

=====

Logged in as: yash_g

1. Book Ticket
2. Show my Tickets
3. Cancel Ticket
4. Logout

Kindly enter your choice: 3

===== CANCEL TICKET =====

-
1. 2026-03-01 | Train 101 - Rajdhani Express | Delhi → Mumbai
 2. 2026-02-28 | Train 103 - Durgam Express | Kolkata → Delhi

Kindly enter ticket number to to cancel the ticket (0 to go back)

Are you sure you want to cancel PNR 126-104581? (Y/N): y

Releasing confirmed seats...

Days left before journey : 5 day(s)

Refund Percentage : 50%

Refund Amount : Rs. 500

Ticket Cancelled Successfully!
Refund of Rs. 500 will be processed.

=====

Refund will be processed depending on how many days before the journey date the ticket is cancelled.

```
3
4 # Add refund entry in payments.txt
5 if [ "$days_left" -gt 20 ]; then
6     refund_percent=100
7 elif [ "$days_left" -ge 8 ]; then
8     refund_percent=80
9 elif [ "$days_left" -ge 1 ]; then
10    refund_percent=50
11 else
12    refund_percent=0
13 fi
14 refund_amount=$((fare * refund_percent / 100))
15
```

```
9 101|2026-03-01|SL|S1|1|LB|Available
10 101|2026-03-01|SL|S1|35|UB|Available
11 101|2026-03-01|3AC|IR1|11|R|Available
```

These two seats were booked during last time; but when canceled here, they will be shown as Available again in seats.txt.